Adaptive Portfolio Decision Engine (APDE)

Expected Returns

From ML Prediction Engine
Annualized multi-horizon

Covariance Matrix

Historical returns.cov() x 252
252-day lookback

Current Weights

Portfolio holdings
+ CDE candidates

Multi-Objective Utility Maximization

lambda = risk aversion (1 to 5 scale)

$$\max_{\Delta \mathbf{w}} U = \mathbf{w}_{new}^T \hat{\mathbf{r}} - \frac{\lambda}{2} \mathbf{w}_{new}^T \Sigma \mathbf{w}_{new} - \gamma \sum |\Delta w_i|$$

gamma = 0.01 transaction cost

Operational Constraints

1. Turnover Cap

Sum|dw|/2 <= C_realloc

User-defined max %

2. No Short-Selling

w_i + dw_i >= 0

Long-only

3. Budget

Sum(w) = 1.0

Fully invested

4. Sector Limits

Sum(w_sec) <= S_max

Tech<=85%

SLSQP Solver (scipy.optimize)

Sequential Least Squares Programming
Bounded multivariate optimization

Optimal dw Vector

Reductions: stocks to sell
Additions: stocks to buy

Decision Scoring (0-100)

Return Improvement (40pts) | Risk (30pts)
Diversification (15pts) | Confidence (15pts)

Decision Card Generation

Reallocation % | Capital Amount | 4-D Explainability
-> User Governance (Accept / Reject) -> Paper Trading